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Adapter assembly for use with a wellbore tool string

Abstract

An adapter assembly may include a tandem seal adapter (TSA) and a collar. The TSA may include a TSA body extending along an axial direction. The collar may include a collar body, a first collar coupling, and a second collar coupling. The collar body may be a substantially annular shape extending in the axial direction. The collar may be outward from the TSA in a radial direction. The first collar coupling and the second collar coupling axially displaced from the first collar coupling may be provided on an interior surface of the collar body. The TSA body and the collar body may overlap in the axial direction. The first collar coupling, second collar coupling, and the TSA may define a first space respectively configured to receive an end of a first housing of a first wellbore tool and an end of a second housing of a second wellbore tool.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/545,147 filed Dec. 8, 2021, which is a continuation of U.S. patent application Ser. No. 17/181,280 filed Feb. 22, 2021, which is a continuation-in-part of U.S. Design patent application No. 29/735,905, filed May 26, 2020 and claims priority to U.S. Provisional Application No. 62/992,643 filed Mar. 20, 2020, the contents of each of which are incorporated herein by reference.

BACKGROUND
(1) Wellbore tools used in oil and gas operations are often sent down a wellbore in tool strings including multiple discrete wellbore tools, or modules, connected together to consolidate different or multiple wellbore operations into a single "run," or process of sending wellbore tools downhole to perform one or more operations. This approach contributes to time and cost savings because preparing and deploying a wellbore tool into a wellbore and pumping, with fluid under hydraulic pressure, the wellbore tool to a particular location in a wellbore (that may be a mile or more under the ground) requires a great deal of time, energy, and manpower. Additional time, manpower, and costs are required to conduct the operation and remove the spent wellbore tool(s) from the wellbore.
(2) Wellbore tools may include, without limitation, perforating guns, puncher guns, logging tools, jet cutters, plugs, frac plugs, bridge plugs, setting tools, self-setting bridge plugs, self-setting frac plugs, mapping/positioning/orientating tools, bailer/dump bailer tools and ballistic tools. Many of these wellbore tools contain sensitive or powerful explosives because many wellbore tools are ballistically (i.e., explosively) actuated or perform ballistic operations within the wellbore. Additionally, certain wellbore tools may include sensitive electronic control components and connections that control various operations of the wellbore tool. Explosives, control systems, and other components of wellbore tools may be sensitive to conditions within the wellbore including the high pressures and temperatures, fluids, debris, etc. In addition, wellbore tools that have explosive activity may generate tremendous amounts of ballistic and gas pressures within the wellbore tool itself. Accordingly, to ensure the integrity and proper operation of wellbore tools connected together as part of the tool string, connections between adjacent wellbore tools within the tool string may not only connect adjacent wellbore tools in the tool string, they may, in many cases, seal internal components of the wellbore tools from the wellbore conditions and pressure isolate adjacent modules against ballistic forces.
(3) A tandem seal adapter (TSA) is a known connector often used for accomplishing the functions of a connector as described above, and in particular for connecting adjacent perforating gun modules. A perforating gun is an exemplary, though not limiting, wellbore tool that may include many of the features and challenges described above. A perforating gun carries explosive charges into the wellbore to perform perforating operations by which the shaped charges are detonated in a manner that produces perforations in a surrounding geological hydrocarbon formation from which oil and gas may be recovered. Conventional perforating guns often include electric componentry to control positioning and detonation of the explosive charges.

(4) In conventional systems, problems may arise in that the mechanical coupling between consecutive wellbore tools has insufficient strength. Additionally, conventional connectors may undesirably increase the length of the wellbore tool string. For example, a conventional connector may include both sealing elements and mechanical coupling components on the same part. However, as the sealing elements and coupling components must be axially separated, this increases the overall axial length of the connector, which in turn increases the length of the tool string.

(5) Accordingly, it may be desirable to develop a tandem seal adapter, adapter assembly, and wellbore tool string that helps to strength mechanical coupling between components, shortens the length of the tool string, and may be produced more efficiently and inexpensively.

BRIEF DESCRIPTION

(6) An exemplary embodiment of an adapter assembly for use with a wellbore tool string may include a tandem seal adapter (TSA) and a collar. The TSA may include a TSA body extending along an axial direction. The collar may include a collar body, a first collar coupling, and a second collar coupling. The collar body may be formed in a substantially annular shape and extend in the axial direction. The collar may be provided outward from the TSA in a radial direction substantially perpendicular to the axial direction. The first collar coupling and the second collar coupling may be provided on an interior surface of the collar body. The second collar coupling may be axially displaced from the first collar coupling. The TSA body and the collar body overlap in the axial direction. The first collar coupling and the TSA may define a first space configured to receive an end of a first housing of a first wellbore tool. The second collar coupling and the TSA may define a second space configured to receive an end of a second housing of a second wellbore tool.

(7) An exemplary embodiment of an adapter assembly for use with a wellbore tool string may include a tandem seal adapter (TSA) and a collar. The TSA may include a TSA body extending along an axial direction. The collar may include a collar body, a first collar coupling, and a second collar coupling. The collar body may be formed in a substantially annular shape and extend in the axial direction. The collar may be provided outward from the TSA in a radial direction substantially perpendicular to the axial direction. The first collar coupling and the second collar coupling may be provided on an interior surface of the collar body. The second collar coupling may be axially displaced from the first collar coupling. The TSA body and the collar body overlap in the axial direction. an outer diameter of the collar decreases in a direction from a center of the collar in the axial direction to a first end of the collar in the axial direction. An outer diameter of the collar may decrease in a direction from the center of the collar in the axial direction to a second end of the collar in the axial direction.

(8) An exemplary embodiment of an adapter assembly for use with a wellbore tool string may include a tandem seal adapter and a collar. The TSA may include a TSA body extending along an axial direction, a first seal and a second seal provided to a first side of a center of the TSA body in the axial direction, and a third seal and a fourth seal provided to a second side of the center of the TSA body in the axial direction. The collar may include a collar body, a first collar coupling, and a second collar coupling. The collar body may be formed in a substantially annular shape and extend in the axial direction. The collar may be provided outward from the TSA in a radial direction substantially perpendicular to the axial direction. The first collar coupling provided on an interior surface of the collar body. The second collar coupling providing on the interior surface of the collar body and axially displaced from the first collar coupling. The TSA body and the collar body overlap in the axial direction. The first seal and the second seal may overlap with the first collar coupling in the axial direction. The third seal and the fourth seal overlap with the second collar coupling in the axial direction.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

(1) A more particular description will be rendered by reference to exemplary embodiments that are illustrated in the accompanying figures. Understanding that these drawings depict exemplary embodiments and do not limit the scope of this disclosure, the exemplary embodiments will be described and explained with additional specificity and detail through the use of the accompanying drawings in which:

(2) FIG. **1** is a cross-section view of a wellbore tool string according to an exemplary embodiment;

(3) FIG. **2**A is a cross-section view of a tandem seal adapter according to an exemplary embodiment;

(4) FIG. **2**B is a cross-section view of a tandem seal adapter according to an exemplary embodiment;

(5) FIG. **3** is a cross-section view of a collar according to an exemplary embodiment;

(6) FIG. **4**A is an enlarged cross-section view of a wellbore tool string according to an exemplary embodiment;

(7) FIG. **4**B is an enlarged cross-section view of a wellbore tool string according to an exemplary embodiment;

(8) FIG. **5** is an enlarged cross-section view of an adapter assembly according to an exemplary embodiment;

(9) FIG. **6** is an enlarged cross-section view of a wellbore tool housing according to an exemplary embodiment;

(10) FIG. **7** is an enlarged cross-section view of a wellbore tool string according to an exemplary embodiment;

(11) FIG. **8** is a flowchart illustrating a method of using a wellbore tool string according to an exemplary embodiment; and

(12) FIG. **9** is a flowchart illustrating a method of assembling a wellbore tool string according to an exemplary embodiment.

(13) Various features, aspects, and advantages of the exemplary embodiments will become more apparent from the following detailed description, along with the accompanying drawings in which like numerals represent like components throughout the figures and detailed description. The various described features are not necessarily drawn to scale in the drawings but are drawn to emphasize specific features relevant to some embodiments.

DETAILED DESCRIPTION

(14) Reference will now be made in detail to various exemplary embodiments. Each example is provided by way of explanation and is not meant as a limitation and does not constitute a definition of all possible embodiments.

(15) The present disclosure may use the term "substantially" in phrases including, but not limited to, "substantially annular shape," "substantially parallel," and "substantially perpendicular," hereinafter summarized as "substantially [x]." In the context of this disclosure, the phrase "substantially [x]" is meant to include both "precisely [x]" and deviations from "precisely [x]" such that the structure would function, from the perspective of one of ordinary skill in the art, in the same way as if it were "precisely [x]." The word "substantially" is not itself limiting but would be readily understood by a person of ordinary skill in the art in view of the exemplary embodiments described in this disclosure and shown in the figures.

(16) FIG. **1** shows an exemplary embodiment of an adapter assembly **108** for use in a wellbore tool string **106**. The wellbore tool string **106** may include a first wellbore tool **118** having a first housing **120**, a second wellbore tool **122** having a second housing **124**, and the adapter assembly **108**. The adapter assembly **108** may be configured to mechanically and electrically couple the first wellbore tool **118** to the second wellbore tool **122**. Additionally, the adapter assembly **108** may be configured to sealingly isolate the first wellbore tool **118** from the second wellbore tool **122** with regard to

fluid and pressure penetration. Additionally, the adapter assembly **108** may be configured to sealingly isolate the first wellbore tool **118** and the second wellbore tool **122** from fluids and pressure exterior to the wellbore tool string **106**.

(17) The adapter assembly **108** may include a tandem seal adapter (TSA **110**) comprising a TSA body **112**. The TSA body **112** may extend along an axial direction **102**. In an exemplary embodiment, the TSA body **112** may have a total length of 1 inch or less in the axial direction **102**. The adapter assembly **108** may further include a collar **114**. The collar **114** may include a collar body **116** formed in a substantially annular shape. The collar body **116** may extend in the axial direction **102**. The collar **114** may be provided outward from the TSA **110** in a radial direction **104**, the radial direction **104** being substantially perpendicular to the axial direction **102**. The TSA **110** and the collar **114** may overlap in the axial direction **102**.

(18) FIG. **2**A and FIG. **2**B illustrate an exemplary embodiment of the TSA **110**. The TSA **110** may include a TSA rib **204** extending radially outward from the TSA body **112** in the radial direction **104**. Further details of the TSA rib **204** will be discussed herein with reference to FIG. **5** and FIG. **7**.

(19) As seen in FIG. **2**A, the TSA **110** may include sealing elements provided on an outer surface **202** of the TSA body **112**. In the example shown in FIG. **2**A, the sealing elements may include a first seal **206**, a second seal **208**, a third seal **210**, and a fourth seal **212**. However, it will be understood that the specific number of seals may be variable to suit a particular application. In an exemplary embodiment, the first seal **206**, the second seal **208**, the third seal **210**, and the fourth seal **212** may be o-rings. The first seal **206**, the second seal **208**, the third seal **210**, and the fourth seal **212** may be respectively provided within a First TSA seal groove **218**, a Second TSA seal groove **220**, a Third TSA seal groove **222**, and a Fourth TSA seal groove **224** formed in the outer surface **202** of the TSA body **112** (see FIG. **2**B).

(20) As seen in FIG. **2**A, the first seal **206** and the third seal **210** may be provided to a first side of a TSA center **214** (approximate position of the TSA center **214** is shown by the broken line in FIG. **2**A), and the second seal **208** and the fourth seal **212** may be provided to a second side of the TSA center **214**.

(21) FIG. **2**B shows an exemplary embodiment of the TSA **110**, which may further include a bore **216** extending through the TSA body **112**. Returning to FIG. **2**A, a bulkhead **226** may be provided within the bore **216**. Exemplary embodiments of the bulkhead **226** are described in U.S. patent application Ser. No. 16/819,270, filed Mar. 16, 2020, which is herein incorporated by reference to the extent that it does not conflict with the present application. The bulkhead **226** may sealingly isolate the first wellbore tool **118** from the second wellbore tool **122**, for example via bulkhead seals **228***a*, **228***b*, **228***c*, **228***d*.

(22) The bulkhead **226** may include a first electrical contact **230** and a second electrical contact **232** that are in electrical communication through an interior of the bulkhead **226**. The first electrical contact **230** is configured to contact a component within the first wellbore tools **118**, and the second electrical contact **232** is configured to contact a component with the second wellbore tool **122**, thereby providing electrical communication between the first wellbore tool **118** and the second wellbore tool **122** through the TSA **110**.

(23) The bulkhead **226** may be retained in the bore **216** by abutting with an interior shoulder **234** of the TSA body **112** at a first end. A retainer nut **236** may be used to retain the bulkhead **226** within the bore **216** at a second end. The retainer nut **236** may be threadedly engaged with the TSA body **112**. It will be understood that other structures may be used in place of the retainer nut **236**, such as a C-clip or a retainer ring.

(24) FIG. **3** illustrates an exemplary embodiment of the collar **114**. The collar **114** may include a collar rib **302** extending radially inward from the collar body **116** in the radial direction **104**. The collar **114** may further include a first collar coupling **306** and a second collar coupling **308**. In an exemplary embodiment, the first collar coupling **306** and the second collar coupling **308** may be

provided on an interior surface of the collar body **116**. The first collar coupling **306** and the second collar coupling **308** may be embodied as threads formed on the interior surface of the collar body **116**. The first collar coupling **306** may be provided to a first side of a collar center **304** in the axial direction **102** (approximate location of the collar center **304** is indicated by the broken line). The second collar coupling **308** may be provided to a second side of the collar center **304** in the axial direction **102**.

(25) In an exemplary embodiment, the collar **114** may have a maximum outer diameter of about 3.5 inches at the collar center **304**. The collar may further include a first sloped portion **310** and a second sloped portion **312** where an outer diameter of the collar **114** decreases as distance from the collar center **304** increases. This may help to provide a tapered profile at ends of the collar **114** that help to prevent or reduce friction, shock, and damage in the event of impact with a wellbore casing during a pump-down operation.

(26) Additionally, as the outer diameter of the collar **114** may be larger than an outer diameter of connected wellbore tools, the collar **114** may help to prevent contact between the wellbore tools and the wellbore casing, thereby reducing the chance of contact and damage to both the wellbore tools and the wellbore casing. Additionally, larger diameter of the collar **114** may help to centralize wellbore tools within the wellbore, thereby resulting in more consistent diameters of perforations into the surrounding formations.

(27) FIG. **4**A is an enlarged cross-section view showing adapter assembly **108**. In an exemplary embodiment, the TSA rib **204** and the collar rib **302** may overlap in the axial direction **102**. Additionally, the TSA rib **204** and the collar rib **302** may overlap in the radial direction **104**. The first seal **206** and the third seal **210** may overlap with the first collar coupling **306** in the axial direction **102**, and the second seal **208** and the fourth seal **212** may overlap with the second collar coupling **308** in the axial direction **102**. As further seen in FIG. **4**A, the first housing **120** may be provided between the first seal **206** and the first collar coupling **306** in the radial direction **104**. Additionally, the second housing **124** may be provided between the second seal **208** and the second collar coupling **308** in the radial direction **104**.

(28) As further seen in FIG. **4**A, a portion of the first housing **120** may be provided between the TSA body **112** and the collar body **116** in the radial direction **104**. The first housing **120** of the first wellbore tool **118** may abut one or more of the TSA rib **204** and the collar rib **302**. Similarly, a portion of the second housing **124** may be provided between the TSA body **112** and the collar body **116** in the radial direction **104**. The second housing **124** of the second wellbore tool **122** may abut one or more of the TSA rib **204** and the collar rib **302**. The first housing **120** may include a first tool coupling **402** provided on an outer surface of the first housing **120**. Similarly, the second housing **124** may include a second tool coupling **404** provided on an outer surface of the second housing **124**. In an exemplary embodiment, the first tool coupling **402** and the second tool coupling **404** may be threads respectively formed on the outer surfaces of the first housing **120** and the second housing **124**. The first tool coupling **402** may be configured to engage with the first collar coupling **306** to mechanically couple the first housing **120** to the collar body **116** of the collar **114**. Similarly, the second tool coupling **404** may be configured to engage with the second collar coupling **308** to mechanically couple the second housing **124** to the collar body **116** of the collar **114**. When the first tool coupling **402** is engaged with the first collar coupling **306**, the first seal **206** and the third seal **210** may overlap with both the first tool coupling **402** and the first collar coupling **306** in the axial direction **102**. Similarly, when the second tool coupling **404** is engaged with the second collar coupling **308**, the second seal **208** and the fourth seal **212** may overlap with both the second tool coupling **404** and the second collar coupling **308** in the axial direction **102**.

(29) Using the adapter assembly **108** to connect the first wellbore tool **118** and the second wellbore tool **122** (see FIG. **1**) may help to decrease the overall length of the wellbore tool string **106**. For example, in an exemplary embodiment, the adapter assembly **108** includes separate pieces such as the TSA **110** and the collar **114**. By providing the sealing elements (such as the first seal **206**, the

second seal **208**, the third seal **210**, and the fourth seal **212**) on the TSA **110** and the coupling elements (such as the first collar coupling **306** and the second collar coupling **308**) on the collar **114**, the sealing elements and the coupling elements can overlap in the axial direction **102**, instead of having to be axially displaced from each other. Accordingly, the overall length of the adapter assembly **108** may be shortened compared with conventional devices. This may allow for shorting of the entire wellbore tool string **106**.

(30) FIG. **4**B shows the relative dimensions of exemplary embodiments of the TSA body **112**, the collar **114**, and the first housing **120**. A TSA body diameter **406** in the radial direction **104** may be smaller than an inner collar diameter **408** in the radial direction **104**. An outer collar diameter **410**, i.e., an outer adapter assembly diameter, in the radial direction **104** may be larger than an outer tool diameter **412**, i.e., an outer first housing diameter, in the radial direction **104**. In an exemplary embodiment, the outer collar diameter **410** may be 3.5 inches and the outer tool diameter **412** may be 3.125 inches.

(31) The relative dimensions of the outer collar diameter **410** and the outer tool diameter **412** may help to improve efficiency during pump-down operations of the wellbore tool string **106**. For example, because the outer collar diameter **410** is larger than the outer tool diameter **412**, the surface area of the wellbore tool string **106** in contact with an inner surface of the wellbore is reduced, thereby reducing surface friction that may acting in opposition to the pump-down operation, especially in applications where the wellbore has a horizontal component with respect to gravity. Further, the differential between the outer collar diameter **410** and the outer tool diameter **412** provides an increased cross-sectional surface area for wellbore fluid to press against during a pump-down operation. In an exemplary embodiment in which the wellbore tools are perforating guns, the outer tool diameter **412** may increase and approach the outer collar diameter **410** following firing of the perforation guns due to gun swell. This may reduce the cross-sectional surface area to facilitate withdrawal of the wellbore tool string **106** from the wellbore.

(32) FIG. **5** shows an enlarged cross-section view of an exemplary embodiment of the TSA rib **204** and the collar rib **302**. As seen in FIG. **5**, the TSA rib **204** has a stepped profile when viewed in cross-section, in other words, when viewed along a plane intersecting with a central axis **238** of the TSA **110**. For example, the TSA rib **204** may include a first TSA rib wall **502** extending radially outward from the TSA body **112** in the radial direction **104**. The TSA rib **204** may further include a second TSA rib wall **504** extending radially outward from the TSA body **112**, with the second TSA rib wall **504** being spaced apart from the first TSA rib wall **502** in the axial direction **102**. The TSA rib **204** may further include a first TSA rib step surface **506** extending from the first TSA rib wall **502** in the axial direction **102** toward the second TSA rib wall **504**. The TSA rib **204** may further include a second TSA rib step surface **508** extending from the second TSA rib wall **504** in the axial direction **102** toward the first TSA rib wall **502**. The first TSA rib step surface **506** and the second TSA rib step surface **508** may be spaced apart in the radial direction **104**. The TSA rib **204** may further include a third TSA rib wall **510** extending in the radial direction **104** from the first TSA rib step surface **506** to the second TSA rib step surface **508**.

(33) As further seen in FIG. **5**, the collar rib **302** and the third TSA rib wall **510** may overlap in the radial direction **104**, and the collar rib **302** and the first TSA rib step surface **506** may overlap in the axial direction **102**. The collar rib **302** may abut one or more of the first TSA rib step surface **506** and the third TSA rib wall **510**. The second TSA rib step surface **508**, the collar rib **302**, and the collar body **116** may define a recess **512** for receiving a portion of the first housing **120**.

(34) FIG. **6** shows an enlarged cross-section view of the first housing **120** according to an exemplary embodiment. The first housing **120** may include a first housing rim **602** provided at a first end of the first housing **120**. The first housing rim **602** may be defined in part by a first end surface **604** substantially parallel to the radial direction **104** and a first axial surface **606** extending from the first end surface **604** substantially parallel to the axial direction **102**. The first housing rim **602** may be received in the recess **512** (see FIG. **5**). The first housing **120** may further include a

first tool step surface **608** extending radially inward from the first axial surface **606**. The first axial surface **606** and the first tool step surface **608** may define a tool groove **610** formed in a first housing inner surface **612** of the first housing **120**.
(35) FIG. **7** shows an enlarged cross-section view illustrating the region of the TSA rib **204**, the collar rib **302**, and the first housing rim **602**. As seen in FIG. **7**, at least a portion of the TSA rib **204** is received in the tool groove **610**. The first end surface **604** may abut against the collar rib **302**. One or more of the first axial surface **606** and the first tool step surface **608** may abut against the TSA rib **204**. As can be seen in FIG. **7**, at least a portion of the TSA rib **204** may be interposed between the collar rib **302** and the first tool step surface **608** of the first housing **120** in the axial direction **102**. This may help to lock the TSA **110** in place and prevent movement of the TSA **110** in the axial direction **102**, thereby helping to maintain stable mechanical and electrical connections between the first wellbore tool **118** and the second wellbore tool **122** (see FIG. **1**).
(36) Additionally, as seen in FIG. **7**, the collar body **116** of the collar **114** is provided radially outward from the first housing **120**, with the first housing **120** being interposed between the collar **114** and the TSA body **112**. Similarly, the second housing **124** may be interposed between the **114** and the TSA body **112**. This may help to strengthen the mechanical coupling between the first wellbore tool **118** and the second wellbore tool **122** (see FIG. **1**), thereby reducing the risk of damage, breakage, and/or separation during wellbore operations.
(37) FIG. **8** shows an exemplary embodiment of a method **800** for using a wellbore tool string such as the wellbore tool string **106** (see FIG. **1**). In block **802**, the wellbore tool string **106** is provided. The wellbore tool string **106** may include the first wellbore tool **118**, having the first housing **120**, and the adapter assembly **108**. The adapter assembly **108** may have an adapter diameter in the radial direction **104** (see outer collar diameter **410** in FIG. **4**B) that is larger than the outer tool diameter **412**. In block **804**, the wellbore tool string **106** is inserted into a wellbore. In block **806**, a pump-down operation is performed on the wellbore tool string **106** to position the wellbore tool string **106** at a desired position. For example, the desired position may be a position for firing perforating guns.
(38) As noted above, the differential between the outer collar diameter **410** and the outer tool diameter **412** may be improve efficiency of the pump-down operation by reducing surface area in contact with the wellbore and providing increased cross-sectional surface area for the wellbore fluid to act against.
(39) FIG. **9** shows an exemplary embodiment of a method **900** for assembling a wellbore tool string such as the wellbore tool string **106** (see FIG. **1**). In block **902**, the first housing **120** of the first wellbore tool **118** is provided. In block **904**, the TSA **110** is inserted into the first housing **120** until the TSA rib **204** abuts with the first housing **120**.
(40) In block **906**, the collar **114** is coupled to the first housing **120**. The portion of the TSA **110** protruding from the first housing **120** may be passed through the interior of the collar **114** until the first collar coupling **306** starts to engage with the first tool coupling **402**. In an exemplary embodiment in which the first collar coupling **306** and the first tool coupling **402** are complementary threads, the collar **114** and the first housing **120** may be rotated relative to each other until the collar **114** is securely coupled to the first housing **120**, which may occur when the collar rib **302** abuts one or both of the TSA rib **204** and the first housing **120** (see FIG. **4**A). In this configuration, a portion of the first housing **120** will be positioned between the TSA body **112** and the collar **114** in the radial direction **104**.
(41) In block **908**, the collar **114** is coupled to the second housing **124** of the second wellbore tool **122**. This may be achieved by inserting the second housing **124** into the collar **114** opposite the first housing **120** to engage the second collar coupling **308** and the second tool coupling **404** (see FIG. **4**A). In an exemplary embodiment in which the second collar coupling **308** and the second tool coupling **404** are complementary threads, the collar **114** and the second wellbore tool **122** may be rotated relative to each other until the collar **114** is securely coupled to the second wellbore tool

122, which may occur when the second housing **124** abuts one or both of the TSA rib **204** and the collar rib **302**.

(42) This disclosure, in various embodiments, configurations and aspects, includes components, methods, processes, systems, and/or apparatuses as depicted and described herein, including various embodiments, sub-combinations, and subsets thereof. This disclosure contemplates, in various embodiments, configurations and aspects, the actual or optional use or inclusion of, e.g., components or processes as may be well-known or understood in the art and consistent with this disclosure though not depicted and/or described herein.

(43) The phrases "at least one," "one or more," and "and/or" are open-ended expressions that are both conjunctive and disjunctive in operation. For example, each of the expressions "at least one of A, B and C," "at least one of A, B, or C," "one or more of A, B, and C," "one or more of A, B, or C," and "A, B, and/or C" means A alone, B alone, C alone, A and B together, A and C together, B and C together, or A, B, and C together.

(44) In this specification and the claims that follow, reference will be made to a number of terms that have the following meanings. The terms "a" (or "an") and "the" refer to one or more of that entity, thereby including plural referents unless the context clearly dictates otherwise. As such, the terms "a" (or "an"), "one or more" and "at least one" can be used interchangeably herein. Furthermore, references to "one embodiment", "some embodiments", "an embodiment" and the like are not intended to be interpreted as excluding the existence of additional embodiments that also incorporate the recited features. Approximating language, as used herein throughout the specification and claims, may be applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term such as "about" is not to be limited to the precise value specified. In some instances, the approximating language may correspond to the precision of an instrument for measuring the value. Terms such as "first," "second," "upper," "lower," etc. are used to identify one element from another, and unless otherwise specified are not meant to refer to a particular order or number of elements.

(45) As used herein, the terms "may" and "may be" indicate a possibility of an occurrence within a set of circumstances; a possession of a specified property, characteristic or function; and/or qualify another verb by expressing one or more of an ability, capability, or possibility associated with the qualified verb. Accordingly, usage of "may" and "may be" indicates that a modified term is apparently appropriate, capable, or suitable for an indicated capacity, function, or usage, while taking into account that in some circumstances the modified term may sometimes not be appropriate, capable, or suitable. For example, in some circumstances an event or capacity can be expected, while in other circumstances the event or capacity cannot occur—this distinction is captured by the terms "may" and "may be."

(46) As used in the claims, the word "comprises" and its grammatical variants logically also subtend and include phrases of varying and differing extent such as for example, but not limited thereto, "consisting essentially of" and "consisting of." Where necessary, ranges have been supplied, and those ranges are inclusive of all sub-ranges therebetween. It is to be expected that the appended claims should cover variations in the ranges except where this disclosure makes clear the use of a particular range in certain embodiments.

(47) The terms "determine," "calculate," and "compute," and variations thereof, as used herein, are used interchangeably and include any type of methodology, process, mathematical operation or technique.

(48) This disclosure is presented for purposes of illustration and description. This disclosure is not limited to the form or forms disclosed herein. In the Detailed Description of this disclosure, for example, various features of some exemplary embodiments are grouped together to representatively describe those and other contemplated embodiments, configurations, and aspects, to the extent that including in this disclosure a description of every potential embodiment, variant,

and combination of features is not feasible. Thus, the features of the disclosed embodiments, configurations, and aspects may be combined in alternate embodiments, configurations, and aspects not expressly discussed above. For example, the features recited in the following claims lie in less than all features of a single disclosed embodiment, configuration, or aspect. Thus, the following claims are hereby incorporated into this Detailed Description, with each claim standing on its own as a separate embodiment of this disclosure.
(49) Advances in science and technology may provide variations that are not necessarily express in the terminology of this disclosure although the claims would not necessarily exclude these variations.

Claims

1. An adapter assembly for use with a wellbore tool string, the adapter assembly comprising: a tandem seal adapter (TSA) comprising a TSA body extending along an axial direction; and a collar comprising: a collar body formed in a substantially annular shape and extending in the axial direction, the collar being provided outward from the TSA in a radial direction substantially perpendicular to the axial direction; a first collar coupling provided on an interior surface of the collar body; and a second collar coupling provided on the interior surface of the collar body and axially displaced from the first collar coupling, wherein: the TSA body and the collar body overlap in the axial direction; the first collar coupling and the TSA define a first space configured to receive an end of a first housing of a first wellbore tool; the second collar coupling and the TSA define a second space configured to receive an end of a second housing of a second wellbore tool; the TSA further comprises a TSA rib extending radially outward from the TSA body in the radial direction; the collar further comprises a collar rib extending radially inward from the collar body in the radial direction; and wherein the TSA rib and the collar rib overlap in the radial direction.
2. The adapter assembly of claim 1, wherein; the TSA further comprises a first seal provided on an outer surface of the TSA body; and the first seal overlaps with the first collar coupling in the axial direction.
3. The adapter assembly of claim 2, wherein: the first seal is provided to a first side of a TSA center of the TSA body in the axial direction; the first collar coupling is provided to the first side of a collar center of the collar body in the axial direction; the TSA further comprises a second seal provided on the outer surface of the TSA body to a second side of the TSA center in the axial direction; the second collar coupling is provided to the second side of the collar center in the axial direction; and the second seal overlaps with the second collar coupling in the axial direction.
4. The adapter assembly of claim 3, wherein: the TSA further comprises a third seal provided on the outer surface of the TSA body to the first side of the TSA center in the axial direction; the TSA further comprises a fourth seal provided on the outer surface of the TSA body to the second side of the TSA center in the axial direction; the third seal overlaps with the first collar coupling in the axial direction; and the fourth seal overlaps with the second collar coupling in the axial direction.
5. The adapter assembly of claim 1, wherein the TSA further comprises: a bore extending through the TSA body; and a bulkhead provided within the bore; wherein the bulkhead is configured to provide electrical connectivity through the bore of the TSA body.
6. The adapter assembly of claim 1, wherein the TSA rib and the collar rib overlap in the axial direction.
7. The adapter assembly of claim 1, wherein an outer diameter of the collar decreases in a direction from a center of the collar in the axial direction to a first end of the collar in the axial direction.
8. An adapter assembly for use with a wellbore tool string, the adapter assembly comprising: a tandem seal adapter (TSA) comprising a TSA body extending along an axial direction; and a collar comprising: a collar body formed in a substantially annular shape and extending in the axial direction, the collar being provided outward from the TSA in a radial direction substantially

perpendicular to the axial direction; a first collar coupling provided on an interior surface of the collar body; and a second collar coupling provided on the interior surface of the collar body and axially displaced from the first collar coupling, wherein: the TSA body and the collar body overlap in the axial direction; a first outer diameter of the collar decreases in a direction from a center of the collar in the axial direction to a first end of the collar in the axial direction; a second outer diameter of the collar decreases in a direction from the center of the collar in the axial direction to a second end of the collar in the axial direction; and the TSA further comprises a TSA rib extending radially outward from the TSA body in the radial direction; the collar further comprises a collar rib extending radially inward from the collar body in the radial direction; and wherein the TSA rib and the collar rib overlap in the radial direction.

9. The adapter assembly of claim 8, wherein; the TSA further comprises a first seal provided on an outer surface of the TSA body; and the first seal overlaps with the first collar coupling in the axial direction.

10. The adapter assembly of claim 9, wherein a first housing of a first wellbore tool is provided between the first seal and the first collar coupling in the radial direction.

11. The adapter assembly of claim 9, wherein: the first seal is provided to a first side of a TSA center of the TSA body in the axial direction; the first collar coupling is provided to the first side of a collar center of the collar body in the axial direction; the TSA further comprises a second seal provided on the outer surface of the TSA body to a second side of the TSA center in the axial direction; the second collar coupling is provided to the second side of the collar center in the axial direction; and the second seal overlaps with the second collar coupling in the axial direction.

12. The adapter assembly of claim 11, wherein a second housing of a second wellbore tool is provided between the second seal and the second collar coupling in the radial direction.

13. The adapter assembly of claim 12, wherein: the TSA further comprises a third seal provided on the outer surface of the TSA body to the first side of the TSA center in the axial direction; the TSA further comprises a fourth seal provided on the outer surface of the TSA body to the second side of the TSA center in the axial direction; the third seal overlaps with the first collar coupling in the axial direction; and the fourth seal overlaps with the second collar coupling in the axial direction.

14. The adapter assembly of claim 8, wherein the TSA further comprises: a bore extending through the TSA body; and a bulkhead provided within the bore; wherein the bulkhead is configured to provide electrical connectivity through the bore of the TSA body.

15. The adapter assembly of claim 8, wherein the TSA rib and the collar rib overlap in the axial direction.